AI-Based Parking Management System

Problem Statement

Parking management has become a major challenge in urban areas due to the rapid increase in the number of vehicles. Drivers often spend a lot of time searching for available parking spaces, which leads to traffic congestion, fuel wastage, and delays.

Traditional parking monitoring systems mainly depend on manual supervision and are unable to provide efficient real-time parking analysis. Unauthorized parking and improper vehicle management also create security and operational issues in parking areas.

The proposed system uses multimodal AI techniques to analyze parking surveillance videos, vehicle images, parking-related audio signals, and parking records. The system can monitor parking slot occupancy, detect abnormal parking activities, and improve overall parking management efficiency.

The project aims to provide intelligent real-time parking monitoring and support smart parking infrastructure using AI-based automated parking analysis.

Objectives

• To detect parking slot occupancy using AI techniques.

• To identify unauthorized parking and abnormal parking activities.

• To analyze video, image, audio, and text data using multimodal AI.

• To reduce traffic congestion and improve parking management efficiency.

• To develop an intelligent real-time parking monitoring system.